eg. Fold 1 — X_train. X_test | Hyperparam tuning: X_train — [ ] X_train_new, X_val | then ~~model~~.fit X_train_new; model.predict X_val | X_train, X_Test

$$\theta = \sum_{i=1}^{n} \beta_i \, \phi(x^{(i)})$$

18/03/26

## * Support Vector Machines:-

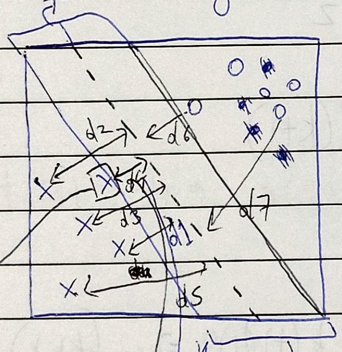
# an algorithm → which will help us to create discriminative classifier.

for linear data —
- Optimal Margin Classifiers / (linear data) ↳ optimal separating hyperplanes
  ↓
- Support Vector Classifier (SVC)
  ↓

linear + non-linear data — Support Vector Machines (SVM)

can be any of the 3 to separate the 2 classes.

2 classes




$X_2$

$X_1$

support vector

# the distance should be less from the line to point or ~~*~~ the data point may be misclassified

# ISV

x = +1

O = −1

margin

∞ a num of lines to classify a data point

· Hyperplane

M1(Blue) = d1 = 3

M2 (Green) → $d_7$ = 1

M3 (Red) = 0.8

M4 (Yellow) = 0

⇒ minimum($d_1, d_2, d_3 \ldots d_4 \ldots d_{11}$)

# ① use validation set for train-test-split for hyperparameter tuning.

↳ X_train ⇒ [ ] } X_train, } validatⁿ ⇒ (X_val)

② loop for each d or hyperparameter:-
model.fit(X_train)
model.predict(X_val)
accuracy score

③ fully train a model
X_train
X_test

classmate Date Page

## * Hyperplane:-

d dim space

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ x_d \end{bmatrix}$$

- hyperplane is an ~~d~~ affine subspace of dimension ~~of~~ d-1
- hyperplane divides the space into 2 halves.

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Hyperplane is defined by

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_d x_d = 0$$

$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ → satisfies this equation then the ~~pt~~ ~~point~~ is on the line.

with $\theta$

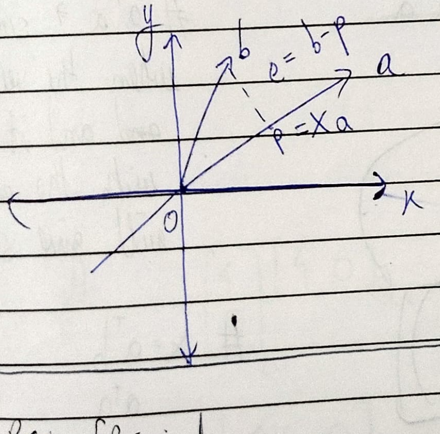
$$\theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2 + \ldots + \theta_d x_d = 0$$

$$\theta = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \vdots \\ \vdots \\ \theta_d \end{bmatrix}$$

26/03

## * Linear algebra Projections:-

x is some part of a till




applicatⁿ:-

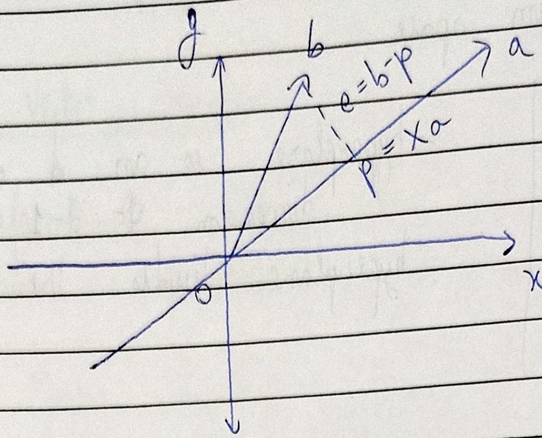
- PCA [Principal
↳ to reduce the ~~dimentionality~~ dimensionality to show the org data in a lower dimension

This is calculated by projectⁿ

- Ax=b (simultaneous eqⁿ) to find approx. solutⁿ, we should project target "b" onto "A".

↳ Gilbert ~~sg~~ .....

e is perpendicular to a or orthogonal.




$a^T e = 0$ ( $\because$ a ⊥ e dot product = 0 )

$p = xa$

$a^T(b - p) = 0$ ~~$p = xa$~~

$a^T(b - xa) = 0$ # $p = xa$

$a^T b - a^T x a = 0$

$a^T b = x a^T a$

# $a^T a$ → scalar value when the vector transpose ~~and~~ and its dot product with the same vector will give scalar value.

$x = \frac{a^T b}{a^T a}$

$p = a x$

$$\boxed{p = a\left(\frac{a^T b}{a^T a}\right)}$$

# $x = \frac{a^T b}{a^T a}$

$p = \frac{a\,a^T b}{a^T a}$ → PM (Project^n Matrix in 1D)

$\boxed{p = PM\, b}$ where $PM = \frac{a\,a^T}{a^T a}$

$$PM = \frac{aa^T}{a^Ta} \quad \begin{matrix} \rightarrow \text{matrix} \\ \rightarrow \text{scalar} \end{matrix} \qquad \cancel{[1\ 2]} \qquad a = \begin{bmatrix}1\\2\end{bmatrix}$$

that's why can't be cancelled. $\quad aa^T \rightarrow \begin{bmatrix}1\\2\end{bmatrix}_{2\times1}\begin{bmatrix}1 & 2\end{bmatrix}_{1\times2}$

In n-dimensional space,

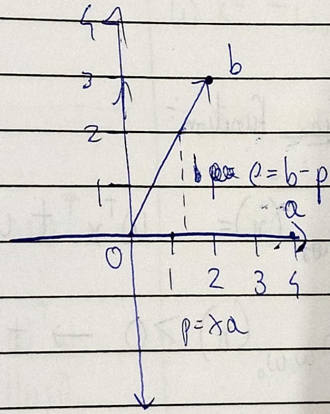
$$\boxed{PM = A(A^TA)^{-1}A^T}$$

let, $b = \begin{bmatrix}2\\3\end{bmatrix}$ $\quad a = \begin{bmatrix}4\\0\end{bmatrix}$ (horizontal line along x-axis)

Find p, i.e. project b ~~onto~~ onto a.

$$p = PM\, b$$

$$PM \Rightarrow \frac{aa^T}{a^Ta}$$




$p = \begin{bmatrix}4\\0\end{bmatrix}_{2\times1}\begin{bmatrix}4 & 0\end{bmatrix}_{1\times2}$ $\qquad p = PM \times b$

$\quad\; p = \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix} \times b$

$\begin{bmatrix}16 & 0\\0 & 0\end{bmatrix}_{2\times2}$

$\rightarrow \begin{bmatrix}4 & 0\end{bmatrix}_{1\times2}\begin{bmatrix}4\\0\end{bmatrix}_{2\times1} \rightarrow 16+0 \rightarrow 16$

$\dfrac{\begin{bmatrix}16 & 0\\0 & 0\end{bmatrix}}{16} \rightarrow \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix} \times \begin{bmatrix}2\\3\end{bmatrix}$

$\rightarrow p = \begin{bmatrix}2\\0\end{bmatrix}$